PSM REPORT FORMATTING AND LATEX GUIDELINES

INSTITUTE OF POSTGRADUATE STUDIES

BACHELOR OF COMPUTER SCIENCE

UNIVERSITI MALAYSIA PAHANG

PSM REPORT FORMATTING AND LATEX GUIDELINES

INSTITUTE OF POSTGRADUATE STUDIES

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ABSTRAK

Dokumen ini menggabungkan spesifikasi pemformatan daripada Panduan Tesis Universiti Malaysia Pahang (2019) dengan garis panduan isi kandungan laporan Projek Sarjana Muda (PSM). Ia juga menerangkan cara menggunakan kelas LaTeX `umpthesis.cls` dan templat `main.tex` untuk menghasilkan laporan yang mematuhi format tersebut. Bab-bab utama merangkumi spesifikasi teknikal dan pemformatan, susun atur isi laporan PSM, penggunaan kelas LaTeX langkah demi langkah, serta senarai semak dan isu-isu yang diketahui.

ABSTRACT

This document combines the formatting specifications from the Universiti Malaysia Pahang Thesis Guide (2019) with the chapter-by-chapter content outline for the Projek Sarjana Muda (PSM) report. It also explains how to use the `umpthesis.cls` LaTeX class and the `main.tex` template to produce a report that follows the required format. The main chapters cover technical and formatting specifications, PSM report structure and content, step-by-step use of the LaTeX template, and a submission checklist with known issues.

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CHAPTER 1

INTRODUCTION

1.1 Purpose

This guideline is a practical reference for preparing the Projek Sarjana Muda (PSM) report at Universiti Malaysia Pahang (UMP). It combines two sources:

1. the **UMP Thesis Guide** (November 2019), which defines the technical and formatting specifications; and
2. the **PSM content guideline**, which defines what should be written in each chapter of the report.

The formatting rules in this document follow the *Publication Manual of the American Psychological Association*, Sixth Edition (APA 6th), as required by the UMP Thesis Guide [Universiti Malaysia Pahang, 2019]. Where a rule can be produced with LaTeX, this document shows the exact command or setting, based on the `umpthesis.cls` class and the `main.tex` template [Ramli, 2017].

1.2 Definition of Terms

Typeface or font All styles such as bold and italics, and the acceptable fonts such as Times New Roman, Arial, and Courier.

Text The main body of the report. It excludes the cover page, preliminaries, tables, figures, references, and appendices.

Preliminaries The front section of the report: title page, declaration pages, dedication (optional), acknowledgements, abstrak, abstract, table of contents, list of tables, list of figures, and list of abbreviations or symbols.

Table Verbal, numeric, or graphical information arranged systematically in rows and columns.

Figure Any information that is not tabulated: illustrations, graphs, charts, diagrams, photographs, or maps.

Equation A mathematical or chemical expression that is part of the text. Equations must be centrally aligned.

Language The report may be written in English or Bahasa Melayu. Other languages require approval from the University Senate.

Appendix Supplementary material that is relevant to the text but would disrupt the flow if included in the main body.

1.3 How to Use This Guideline

- **Chapter 2** lists the formatting rules and the matching LaTeX setting.
- **Chapter 3** lists the required content of each PSM report chapter.
- **Chapter 4** gives step-by-step instructions for compiling a report with the `main.tex` template and `umpthesis.cls`.
- **Chapter 5** provides a submission checklist and documents known differences between the class and the 2019 guide.

CHAPTER 2

TECHNICAL AND FORMATTING SPECIFICATIONS

This chapter summarises the formatting rules from the UMP Thesis Guide 2019 [Universiti Malaysia Pahang, 2019]. Each subsection ends with a **LaTeX note** that shows how the rule is applied when using the `main.tex` template with `umpthesis.cls`.

2.1 Paper Specifications

Table 2.1 lists the paper requirements.

Table 2.1 Paper specifications

Element	Specification
Size	A4 (210 $\times$ 297 mm) for both soft copy and hard copy.
Quality	Original: simile paper, minimum 80 gsm. Duplicates: good quality photocopy or offset paper. Isograph and carbon copies are not acceptable.
Colour	White paper only.
Printing	One side of the paper only. The original must be printed with a laser printer.
Binding	Final submission: three hard-bound copies (one original and two duplicates). Master's cover is black; PhD cover is maroon. For undergraduate PSM reports, cover colour is at the faculty's discretion.

LaTeX note. The class loads `a4paper` and `oneside`. Print one-sided from the PDF viewer.

2.2 Typeface or Font

All text, tables, and graphs must be typeset, not handwritten. Use **Times New Roman 12 pt** throughout the report, including headings and page numbers. Table entries

may use Times New Roman 11 pt or smaller, as long as they remain legible. Equations must be produced with an equation editor, including subscripts and superscripts.

LaTeX note. The class loads `mathptmx` (the standard Times New Roman substitute in LaTeX) at `12pt`:

```
\documentclass[english]{umpthesis}   % 12pt Times-like font
```

2.3 Margin Specifications

All text and graphics must stay within the margins in Table 2.2.

Table 2.2 Margin specifications for A4 paper

Element	Portrait	Landscape
Top edge	25 mm	35 mm
Right side	25 mm	25 mm
Left side	35 mm	25 mm
Bottom edge	25 mm	25 mm

LaTeX note. The class itself does not set the body margins, so the main file must set them explicitly with `geometry`:

```
\geometry{top=25mm,right=25mm,left=35mm,bottom=25mm,includefoot}
```

For landscape pages, use `lscape` or `pdflscape` and set the landscape margins with `\newgeometry` (see Chapter 4).

2.4 Line Spacing

Table 2.3 summarises the spacing rules.

LaTeX note. The class sets the body to approximately 1.5 spacing when `\mainmatter` begins:

```
\linespread{1.435}          % set by umpthesis.cls
```

Table 2.3 Spacing formatting

Line spacing	Applies to
25 mm	Spacing between the upper edge of the page and the chapter number.
1.5-line spacing	Between the chapter number and chapter title; between the title and the first line of text; between the last line of text and a subsection title; between a subsection title and the first line of text; between paragraphs; between the last line of text and a table or figure; list of tables, figures, symbols, and abbreviations; general body text.
Single (1.0)	Acknowledgements; abstract and its translation; long title entries in headings; long tables and table entries; long quotations, notes, and multiline captions; footnotes; bibliographic entries/references.

Acknowledgements, abstract, and dedication environments use `\singlespacing` automatically. Known spacing deviations are listed in Chapter 5.

2.5 Paragraphing

All paragraphs must have the first line indented **12.7 mm** from the left margin. The last paragraph on a page must have at least two lines; otherwise it must begin on the next page. The last word on a page must not be split with a hyphen.

LaTeX note. The class sets the indent and penalties:

```
\setlength{\parindent}{0.5in}   % 12.7 mm, set by umpthesis.cls
\widowpenalty10000
\clubpenalty10000
```

2.6 Pagination and Numbering

Page numbers use the same Times New Roman font as the text, are centred, and sit within **20 mm** of the bottom edge. Preliminaries are numbered in lowercase Roman numerals (i, ii, iii). The text uses Arabic numerals (1, 2, 3). The first page of the preliminaries is counted but its number is not printed. Letter suffixes (10a, 10b) are not allowed.

LaTeX note. The class uses `\frontmatter` and `\mainmatter` to switch numbering:

```
\frontmatter   % roman page numbers for preliminaries
\mainmatter    % arabic page numbers for the text
```

Known issue: the class currently hides the page number on the first page of every chapter (see Chapter 5).

2.7 Front Cover, Spine, and Title Page

2.7.1 Front Cover

The front cover uses **gold coloured CAPITAL LETTERS, not bold**, in Times New Roman **20 pt**, in this order:

1. Title of the report (50 mm from the top edge of the cover);
2. Full name of the student (as in identity card or passport);
3. Name of the degree (e.g., BACHELOR OF COMPUTER SCIENCE);
4. University name (UNIVERSITI MALAYSIA PAHANG, 50 mm from the bottom edge).

LaTeX note. Use `\umpCoverPage`. The class renders this page at 18 pt and does not apply the gold colour (see Chapter 5).

2.7.2 Spine

The spine shows, in gold capital letters, Times New Roman 18 pt: student name, abbreviated degree, year of award, and UMP.

LaTeX note. The class does not currently provide a spine command.

2.7.3 Title Page

The title page uses **single spacing and CAPITAL LETTERS**, in this order:

1. Title of the report (same as the cover, 50 mm from the top edge);

2. Full name of the student;

3. Purpose of submission (e.g., *Report submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of . . .*);

4. Faculty/Institute/Centre (Title Case);

5. UNIVERSITI MALAYSIA PAHANG (uppercase);

6. Month and year of award (50 mm from the bottom edge).

LaTeX note. Use `\maketitle`. Known issue: the class title page uses 1.5 spacing instead of single spacing (see Chapter 5).

2.8 Headings and Subheadings

Table 2.4 gives the formatting style for each heading level.

Table 2.4 Formatting of headings and subheadings

Heading level	Formatting
Chapter title (Heading 1)	Centred, boldface, uppercase. Every chapter starts on a new page.
Section title (Heading 2, UMP Level 2)	Left aligned, boldface, Title Case. Paragraph begins below, indented as a normal paragraph with 1.5 spacing. Numbered 2.1. Included in the Table of Contents.
Subsection title (Heading 3, UMP Level 3)	Left aligned, boldface, Title Case. Paragraph begins below, indented with 1.5 spacing. Numbered 2.1.1. Included in the Table of Contents.
Level 4 (UMP Level 4)	Left aligned, boldface, Title Case. Numbered 2.1.1.1. Not included in the Table of Contents.
Level 5 (UMP Level 5)	Left aligned, not numbered, italicized, Title Case. Not included in the Table of Contents.
Heading 7	Roman numbering (i, ii, iii). Use for objectives and scope. Not in Table of Contents.
Heading 8	Arabic numbering (1, 2, 3). Not in Table of Contents.
Heading 9	Bullet numbering. Not in Table of Contents.

LaTeX note. The class maps these levels as follows:

```
\chapter{...}        % Heading 1
\section{...}        % Heading 2 (2.1)
\subsection{...}     % Heading 3 (2.1.1)
\subsubsection{...}  % Heading 4 (2.1.1.1)
\paragraph{...}      % Heading 5 (see known issue)
\begin{enumerate}[label=(\roman*)] ... \end{enumerate}  % Heading 7
\begin{enumerate} ... \end{enumerate}                   % Heading 8
\begin{itemize} ... \end{itemize}                       % Heading 9
```

2.9 Tables

Number tables with Arabic numerals by chapter: the first table in Chapter 3 is `Table 3.1`. Every table needs a short caption. The caption is always placed **above** the table, even for landscape tables. If a table is copied or adapted from a source, cite the source at the bottom:

```
Source: Author (Year).
```

A table must contain only three horizontal lines and no vertical lines: an opening line below the title, a line that closes the column headings, and a closing line at the bottom. Every column must have a heading. Continued tables must repeat the column headings and show *continued* with the table number on subsequent pages.

LaTeX note. Use `booktabs` for the three-line style and place `\caption` before the tabular environment:

```
\begin{table}[htbp]
\caption{Table caption goes above}
\label{tab:example}
\begin{tabularx}{\textwidth}{@{} l X @{}}
\toprule
Heading A & Heading B \\
\midrule
Value & Description \\
```

```
\bottomrule
\end{tabularx}
\end{table}
```

2.10 Figures

Figures (maps, charts, graphs, drawings, diagrams, photographs) must be clear and high quality. Number figures by chapter: `Figure 4.2` is the second figure in Chapter 4. Text inside a figure may use any neat, readable typeface, but the **figure number and caption** must use the same typeface as the body text. The caption is placed **below** the figure and is left aligned. Figures are centred between the left and right margins. If a figure is in landscape orientation, the top of the figure is placed along the bound edge.

LaTeX note. The class configures the caption package for left-aligned captions:

```
\begin{figure}[htbp]
\centering
\includegraphics[width=0.8\textwidth]{figure/example.png}
\caption{Caption goes below the figure}
\label{fig:example}
\end{figure}
```

2.11 Equations

Equations are centred. Equation numbers use Arabic numerals in parentheses on the right margin and are numbered by chapter: `Eq. 4.2` is the second equation in Chapter 4. Break long equations before a convenient plus or minus sign, or just ahead of an equal sign.

LaTeX note. The memoir base class numbers equations by chapter automatically:

```
\begin{equation}
E = mc^{2}
\end{equation}
```

2.12 Citations and References

In-text citations follow the APA 6th author–date style. Every source cited in the text must appear in the References list, and every entry in the list must be cited in the text. The References list starts on a new page with the heading REFERENCES centred, 25 mm from the top. References are not listed at the end of each chapter; they are listed once at the end of the final chapter, in alphabetical and chronological order, with hanging indentation.

LaTeX note. The class does not provide an APA style file. The current template uses `natbib` with `plainnat` (author–year, close to APA but not exact):

```
\usepackage{natbib}
...
\citet{key}   % Author (Year)
\citep{key}   % (Author, Year)
...
\bibliographystyle{plainnat}
\bibliography{bib/mybiblio}
```

For strict APA 6th output, use a dedicated APA bibliography style or `biblatex-apa`. This is a known gap (see Chapter 5).

2.13 Appendices and List of Publications

Appendices hold material that is relevant but too lengthy for the text (raw data, questionnaires, program listings, maps, etc.). They are paginated consecutively with the main text. If there are more than five appendices, include a *List of Appendices* in the Table of Contents. Publications resulting from the study during candidature should be listed clearly in a *List of Publications*.

LaTeX note. Use `\appendix` followed by `\chapter{...}`. The class does not currently define commands for List of Appendices, List of Publications, or List of Symbols/Abbreviations; define them manually or add them to the class.

CHAPTER 3

PSM REPORT CONTENT OUTLINE

This chapter adapts the PSM content guideline [Universiti Malaysia Pahang, 2023] into a clean, numbered outline. It follows the chapter structure of the UMP Thesis Guide [Universiti Malaysia Pahang, 2019].

3.1 Report Structure Overview

The PSM 1 report (proposal stage) normally contains three chapters plus references:

1. Chapter 1: Introduction
2. Chapter 2: Literature Review
3. Chapter 3: Methodology

The final PSM report extends to the five-chapter structure recommended by the UMP Thesis Guide: Introduction, Literature Review, Research Design and Methodology, Results and Discussion, and Conclusion and Recommendations. Students should confirm the exact chapter division with their supervisor.

3.2 Chapter 1: Introduction

3.2.1 Introduction

Describe the project briefly and explain the related issues. Ease the reader into the topic: state the general topic, give background, review related literature briefly, define the terms and scope, and outline the current situation.

3.2.2 Problem Statement

State the current issue that the project seeks to address. A good problem statement answers *who, what, when, where*, and *why*. Recommended elements:

- Describe how things should work.
- Explain the problem and why it matters.
- Support the claim with cited sources.
- Propose a solution.
- Explain the benefits of the proposed solution.
- Conclude by summarising the problem and the solution.

3.2.3 Objective

Objectives are specific actions to be achieved, aligned with the project goal. They should follow the SMART concept: Specific, Measurable, Achievable, Realistic, and Time-framed. Write objectives as a list using roman numbering (i, ii, iii) and active statements. Example:

```
The objectives of this project are:
(i) To study ...
(ii) To develop ...
(iii) To evaluate ...
```

3.2.4 Scope

Explain the boundary of the project and its users: the extent of the work and the parameters within which the study operates. Write the scope in list form (i, ii, iii) and order the items according to the objectives.

3.2.5 Report Organization

Summarise each chapter of the report. Example:

```
This report consists of three chapters. Chapter 1 discusses the
introduction to the project. Chapter 2 explains and discusses ...
```

3.3 Chapter 2: Literature Review

3.3.1 Introduction

Explain the existing problem or solution from previous work. Describe in detail the technique, method, hardware, or technology suitable for the project. The review should answer *What, Why,* and *How*, and comparisons may be shown in tables. State the sources clearly (book, journal, conference, proceedings, thesis, or website). Citations and bibliography follow the UMP Thesis Guide APA style.

3.3.2 Existing Systems or Works

Describe at least three existing systems or works that are relevant to the proposed solution:

1. **System A** — explanation with supporting diagram or interface.
2. **System B** — explanation with supporting diagram or interface.
3. **System C** — explanation with supporting diagram or interface.

3.3.3 Analysis and Comparison of Existing Systems

Compare the three existing systems and review the comparison in terms of advantages and disadvantages. Then discuss how relevant, significant, or impactful the comparison is to the project title.

3.3.4 Summary

Summarise the key findings of Chapter 2.

3.4 Chapter 3: Methodology

3.4.1 Introduction

Discuss the overall approach or framework of the undergraduate project. The chapter should cover the method, technique, or approach used, and discuss the methodology in detail so the project can be accomplished.

3.4.2 Project Management Framework or Methodology

Discuss and justify the selected development or research methodology and its phases. Project-based work may use SDLC (Agile, RAD, etc.); research-based work uses a research framework. Describe the activities in each phase and how they relate to the project.

3.4.3 Project Requirements

State what is needed to complete the project.

- **Project-based:** functional and non-functional requirements, constraints, and limitations. User requirements may come from interviews, surveys, client feedback, or observation of the existing manual process.
- **Research-based:** input, output, process description, constraints and limitations, and case study.

3.4.4 Proposed Design

Present the proposed design based on the requirements.

- **Project-based:** flowchart, context diagram, use case diagram and description, activity diagram, storyboard.
- **Research-based:** pseudocode, algorithm, flowchart, model, or a related design depending on the type of research.

3.4.5 Data Design

Describe the data or assets involved in the project.

- **Project-based:** ERD, database dictionary (primary key, foreign key), assets, and the process of creating the database.
- **Research-based:** dataset description and justification of the selected dataset.

3.4.6 Proof of Initial Concept

For PSM 1, provide at least a proof of concept (POC) that verifies the idea can be executed from a technical point of view.

- **Project-based:** design of the proof of concept with explanation.
- **Research-based:** evidence of early work, experiment, or preliminary findings.

3.4.7 Testing or Validation Plan

Plan the testing or validation process clearly. Examples include selecting data for testing, how testing is implemented, and the measurement or evaluation used (e.g., user acceptance testing, comparison testing). Project-based work describes the testing strategy for the proposed solution; research-based work covers a simple set of inputs, processes, outputs, and comparison of results.

3.4.8 Potential Use of Proposed Solution

Explain the potential use of the proposed solution in real situations or its potential for commercialisation. Discuss the benefits and significance of the project in daily life.

3.4.9 Gantt Chart

Show the project phases from start to completion. For PSM 1, show the estimated duration. The chart should follow the selected development methodology activities, not the report chapters.

3.5 References

References may come from books, journals, conference proceedings, magazines, or websites. Record all references to identify the copyright and originality of the work. Format citations and bibliography according to the UMP Thesis Guide APA style [Universiti Malaysia Pahang, 2019].

3.6 Notes for PSM 2 (Final Report)

For the final report, add:

- **Chapter 4: Results and Discussion** — present and interpret the results, verify that all figures and tables are necessary and appropriate, and discuss the findings in the context of previous work.
- **Chapter 5: Conclusion and Recommendations** — summarise the findings, state the contributions and limitations, and propose future work.

CHAPTER 4

USING THE `MAIN.TEX` TEMPLATE

This chapter shows how to write and compile a PSM report with the `main.tex` template and the `umpthesis.cls` class [Ramli, 2017]. The template follows the structure of the previous UMP proposal template.

4.1 Prerequisites

You need a working TeX Live or MacTeX installation with `pdflatex` and `bibtex`. The class depends on standard packages (memoir, geometry, hyperref, setspace, graphicx, tabularx, caption, enumitem, tocloft, tikz, etc.), which are included in a normal TeX Live installation.

4.2 Project Layout

Keep the report in a single folder:

```
project/
|-- main.tex                % main file (this template)
|-- umpthesis.cls           % the class file
|-- logo-ump-color.png      % UMP logo for declarations
|-- bib/
|   `-- mybiblio.bib        % bibliography database
|-- figure/
|   `-- example.png         % figures
`-- tex/
```

```
|-- abstrak.tex          % Malay abstract
|-- abstract.tex         % English abstract
|-- dedication.tex       % optional
|-- acknowledgement.tex
|-- introduction.tex     % Chapter 1
|-- literature.tex       % Chapter 2
|-- methodologies.tex    % Chapter 3
`-- abbrevia.tex         % list of abbreviations (optional)
```

4.3 Main File Skeleton

The `main.tex` file sets the class, margins, packages, and document profile:

```
\documentclass[english]{umpthesis}

\geometry{top=25mm,right=25mm,left=35mm,bottom=25mm,includefoot}

\usepackage{lipsum}     % dummy text (remove for real report)
\usepackage{natbib}     % author-year citations
\usepackage{booktabs}   % three-line tables

\setcounter{secnumdepth}{3}
\setcounter{tocdepth}{3}
\pagenumbering{roman}
```

4.4 Document Profile Commands

Set the author, school, and thesis information before `\begin{document}`:

```
\author{Your Full Name}
\authorDOB{01/01/1990}
\authorStudID{ABC12345}
\authorNRICPassport{000000-00-0000}
```

```
\umpFaculty{Faculty of Computer Systems \& Software Engineering}
\umpSession{2026/2027}
\umpDegree{Doctor of Philosophy}
\umpField{Computer Science}

\title{Your Research Title}
\titleWrapOne{Your Research Title}
\titleWrapTwo{}
\titleWrapThree{}

\svisorMain{Supervisor Name}
\svisorMainPos{Senior Lecturer}
% \svisorSecond{Co-Supervisor Name}
% \svisorSecondPos{Senior Lecturer}

\submitYear{2026}
\submitMonth{November}
```

4.5 Front Matter Commands

The preliminaries are assembled in this order:

```
\begin{document}

\umpCoverPage              % front cover

\frontmatter               % roman page numbers start here
\maketitle                 % title page

% \umpCopyright{0}{0}{1}   % Declaration of Thesis and Copyright
                            % {CONFIDENTIAL}{RESTRICTED}{OPEN ACCESS}
% \umpLibLetter{RESTRICTED}{2}{Reason 1}{Reason 2}{Reason 3}
```

```
% \umpSvDeclaration          % Supervisor's Declaration
% \umpStudDeclaration        % Student's Declaration

\input{tex/dedication}      % optional
\input{tex/acknowledgement}
\input{tex/abstrak}         % Malay ABSTRAK first
\input{tex/abstract}        % then English ABSTRACT

\tableofcontents
\listoftables
\listoffigures

\mainmatter                 % arabic page numbers start here
```

For a real report, remove the comment marks from the declaration commands that apply. The copyright declaration takes three values, each `1` for selected and `0` for not selected.

4.6 Main Matter Commands

Write each chapter in its own file and `\input` it:

```
\mainmatter

\input{tex/introduction}
\input{tex/literature}
\input{tex/methodologies}
```

Inside a chapter file, use the class heading commands:

```
\chapter{Introduction}
\section{Background}          % 1.1
\subsection{History}          % 1.1.1
```

```
\subsubsection{Detail}        % 1.1.1.1

\begin{enumerate}[label=(\roman*)]  % (i), (ii), (iii)
\item First objective
\item Second objective
\end{enumerate}
```

Cross-reference with `\label` and `\ref`:

```
\section{Background}
\label{sec:background}
See Section \ref{sec:background}.
```

4.7 Tables, Figures, and Equations

4.7.1 Table

Place the caption above the table and use `booktabs` for the required three-line style:

```
\begin{table}[htbp]
\caption{Comparison of existing systems}
\label{tab:comparison}
\begin{tabularx}{\textwidth}{@{} l l l @{}}
\toprule
Feature & System A & System B \\
\midrule
Accuracy & High & Medium \\
Speed & Fast & Slow \\
\bottomrule
\end{tabularx}
\end{table}
```

4.7.2 Figure

Place the caption below the figure:

```
\begin{figure}[htbp]
\centering
\includegraphics[width=0.8\textwidth]{figure/example.png}
\caption{Research methodology}
\label{fig:method}
\end{figure}
```

4.7.3 Equation

Use `equation` for a numbered, centred equation:

```
\begin{equation}
y = mx + c
\end{equation}
```

4.8 References and Bibliography

Create `bib/mybiblio.bib` with BibTeX entries, then cite with `natbib`:

```
\citep{key}   % (Author, Year)
\citet{key}   % Author (Year)
```

At the end of the main file:

```
\bibliographystyle{plainnat}
\bibliography{bib/mybiblio}
```

`plainnat` is author–year and close to APA. For strict APA 6th, replace it with a dedicated APA style such as `biblatex-apa` (requires changing from BibTeX to Biber).

4.9 Compiling the Report

Run the following commands in the project folder:

```
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The extra `pdflatex` passes resolve citations, the table of contents, and cross-references. Check the log for undefined citations or references before submitting:

```
grep -i undefined main.log
```

4.10 Cleaning Generated Files

After the final compile, remove the auxiliary files and keep the sources and PDF:

```
rm -f *.aux *.bbl *.blg *.log *.out *.toc *.lof *.lot
```

CHAPTER 5

SUBMISSION CHECKLIST AND KNOWN ISSUES

5.1 Pre-Submission Checklist

Format

- A4 white paper, printed one side, correct margins (left 35 mm; top, right, bottom 25 mm).
- Times New Roman 12 pt everywhere; table entries may be 11 pt or smaller but legible.
- 1.5 line spacing for body text; single spacing for abstract, acknowledgements, long quotes, and references.
- First line of every paragraph indented 12.7 mm.
- Page numbers centred at the bottom: roman for preliminaries, arabic for text.
- Table captions above tables; figure captions below figures; both numbered by chapter.
- Tables have only three horizontal lines and no vertical lines.
- Equations centred and numbered by chapter.
- Headings follow the levels in Table 2.4.
- References follow APA 6th and are listed once at the end.

Content

- Chapter 1: Introduction, Problem Statement, Objective, Scope, Report Organization.
- Chapter 2: Literature Review with at least three existing systems compared.
- Chapter 3: Methodology covering framework, requirements, design, data, POC, testing plan, and Gantt chart.
- References contain only works cited in the text.

5.2 Known Differences Between `umpthesis.cls` and the 2019 Guide

Table 5.1 lists the differences found when comparing the class with the UMP Thesis Guide 2019. These are documented here as known issues; the class was used unmodified in this document.

5.3 Recommended Class Patches

If strict compliance with the 2019 guide is required, patch `umpthesis.cls` in this priority order:

1. Remove `\aliaspagestyle{chapter}{empty}` so the first page of every chapter shows its page number.
2. Restore chapter page numbers in the Table of Contents.
3. Set body margins inside the class with `\geometry`.
4. Change cover font to 20 pt and title page to single spacing.
5. Add commands for spine, list of symbols/abbreviations, list of appendices, and list of publications.
6. Replace `plainnat` with a strict APA 6th bibliography style.

Table 5.1 Known differences: `umpthesis.cls` vs UMP Thesis Guide 2019

Item	Guide 2019	Class behaviour / workaround
Cover font	Times New Roman 20 pt, gold, not bold	Class uses 18 pt and no gold colour. Keep for now; final cover is usually produced by the printer/binder.
Title page spacing	Single spacing	Class uses 1.5 spacing. Redefine `\maketitle` locally or patch the class.
Page number on first page of chapter	Printed on all text pages	Class hides it (`\aliaspagestyle`). Remove that line in

BIBLIOGRAPHY

Muhammed Ramiza Ramli. umpthesis.cls: Ump thesis latex template, 2017. Version 0.7.

Universiti Malaysia Pahang. *UMP Thesis Guide*. Institute of Postgraduate Studies, Universiti Malaysia Pahang, Kuantan, Malaysia, november 2019 edition, 2019.

Universiti Malaysia Pahang. Guidelines for psm, 2023. Internal document.